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## **An Empirical Study on Multi-Level Momentum and Liquidity Volatility Using Machine Learning**

### **Abstract**

This study investigates the application of machine learning (ML) in portfolio construction, utilizing multi-level momentum factors and volatility-based risk measures to optimize investment performance. Five key factors—one-month momentum (mom1m), twelve-month momentum (mom12m), six-month momentum change (chmom), maximum single-period return (maxret), and return volatility (retvol)—are evaluated using six representative models: Ordinary Least Squares (OLS), Random Forest (RF), Neural Networks (NN), Gradient Boosting Regression Trees (GBRT), LightGBM (Light Gradient Boosting Machine), and XGBoost (eXtreme Gradient Boosting). Using monthly data of 964 companies listed on the Taiwan Stock Exchange from January 1999 to February 2025, this research constructs portfolios via predicted Information Coefficient (IC) signals across thirteen partition strategies. Performance is benchmarked against the Taiwan Weighted Stock Index (TWII). Empirical results demonstrate that nonlinear ML models substantially outperform OLS regression. Specifically, Random Forest and certain Neural Network configurations achieve a 100% winning rate against the market index across all partitions. The Random Forest model attains an out-of-sample R-squared of 0.763 and cumulative log returns exceeding six times the market benchmark over a 134-month evaluation period. These findings confirm the practical value of machine learning in intelligent investment systems and FinTech applications, particularly in capturing nonlinear momentum dynamics and extreme-return signals.

**Keywords:** Machine learning, momentum factors, portfolio construction, Taiwan stock market, FinTech.

## **1 Introduction**

The dynamic evolution of global financial markets, coupled with advancements in information technology, necessitates the continuous refinement of asset pricing and portfolio construction methodologies. Traditional investment approaches, which primarily rely on linear models and fundamental indicators, exhibit limitations in capturing the complex, non-linear dependencies and interaction effects inherent in high-frequency, high-dimensional financial data [1]. This inadequacy has driven the adoption of sophisticated quantitative techniques.

The subsequent sections formally delineate the research motivation, review the relevant literature, and state the objectives and structure of this study.

### **1.1 Research Background and Motivation**

Factor investing constitutes the foundational paradigm in modern quantitative finance, stemming from the Capital Asset Pricing Model (CAPM) [2]. The evolution to multi-factor models, including the integration of the Momentum factor [5], significantly enhanced explanatory power. Nevertheless, the inherent linearity assumption within these traditional models fails to fully account for complex market phenomena, such as non-linear relationships and extreme value risks [1].

The central motivation of this study is to address this methodological gap by applying machine learning (ML) models to forecast stock returns, capitalizing on their superior ability to model non-linear structures [1], [10]. Specifically, we focus on multilayer momentum characteristics and risk indicators, which are demonstrably persistent and influential signals in the equity market [7], [8].

This research analyzes five core factors: one-month momentum (mom1) [7], twelve-month momentum (mom12) [8], change in six-month momentum (chmom) [11], maximum single-period return (maxret) [14], and return volatility (retvol) [9]. The incorporation of these factors ensures comprehensive coverage of short-term inertia, long-term trends, and volatility-related risks [15].

### **1.2 Research Objectives and Methodology**

The primary objective of this research is to rigorously evaluate the predictive performance of non-linear machine learning models against the traditional linear benchmark within the Taiwan stock market [1], [10]. The study aims to construct optimal investment strategies by leveraging the enhanced predictive capability of these

advanced models.

The methodology involves employing six representative ML algorithms: Ordinary Least Squares (OLS), Random Forest (RF) [16], Neural Networks (NNs) with varying depths [13], Gradient Boosting Regression Trees (GBRT), LightGBM [18], and XGBoost [17]. These models span the spectrum of linear, ensemble, and advanced boosting architectures, facilitating a comparative analysis of their efficacy in extracting complex investment signals [1].

The empirical evaluation is structured around a robust cross-sectional analysis using Taiwan stock exchange data. The market universe is divided into thirteen distinct portfolio partitions, ranging in granularity from 10 groups to 964 groups. The performance of portfolios constructed for each model and partition is meticulously assessed against the benchmark Taiwan Weighted Stock Index (TWII).

### **1.3 The Development of Factor Investing**

Since the 1960s, the evolution of asset pricing theories has laid the foundation for modern investment research. The Capital Asset Pricing Model (CAPM) proposed by Sharpe [2] first established a formal linkage between asset risk and expected return, introducing the beta coefficient as a measure of systematic risk relative to the market portfolio. The model posits that investors who bear higher systematic risk should be compensated with higher expected returns, forming the cornerstone of modern asset pricing theory.

Subsequent empirical evidence revealed the existence of numerous anomalies that could not be explained by a single market factor. Haugen and Heins [9] identified the Low Volatility Anomaly, suggesting that stocks with lower volatility often yield higher risk-adjusted returns, thereby challenging the traditional "high-risk, high-return" assumption. Later, Fama and French [3] introduced the Three-Factor Model, which incorporated firm size (Size) and book-to-market ratio (Value) as additional explanatory variables beyond the market factor, substantially improving explanatory power. The model was later expanded into a Five-Factor Model [4] by adding profitability and investment factors, offering a more comprehensive view of return generation mechanisms.

Carhart [5] further extended this framework by incorporating the Momentum factor, which captures the tendency of stocks with strong (or weak) recent performance to continue their trends over the short term. The discovery of momentum effects

significantly enhanced the explanatory and practical utility of multifactor models, making them central to quantitative investment strategies [15].

Harvey, Liu, and Zhu [6] reviewed the growing literature and noted that with increasing data availability and computational capacity, more than 240 potential return predictors—commonly known as "factors"—have been identified in academic and practical contexts, with projections suggesting that this number could exceed 500 in the future. Nevertheless, only a limited subset—such as low volatility, value, size, momentum, profitability, and quality—has demonstrated stable explanatory power across different markets and time periods [6].

In summary, the evolution of factor investing has progressed from single-factor linear models to multifactor and nonlinear frameworks. With advancements in financial technology and data analytics, traditional linear approaches are increasingly limited in capturing complex market structures, paving the way for the integration of machine learning techniques into modern asset pricing research [1], [10].

#### **1.4 Applications of Machine Learning in Factor Investing**

In recent years, with the rapid growth of financial technology (FinTech), machine learning (ML) has emerged as a pivotal analytical tool in asset pricing and investment decision-making. Gu, Kelly, and Xiu [1], in *The Review of Financial Studies*, demonstrated that ML algorithms significantly improve the predictive accuracy of asset returns, particularly in capturing nonlinear relationships and interaction effects among factors—areas where traditional linear models often underperform. Similar evidence has been reported in the Chinese equity market by Leippold, Wang, and Zhou [10].

The advantages of machine learning can be summarized in two key dimensions. First, ML algorithms are highly capable of processing large-scale, high-dimensional financial data, effectively extracting stable predictive signals from noisy environments [1]. Second, through automated feature learning and model training, ML models can identify latent structures, enhancing both robustness and generalization performance [13]. These properties make machine learning a powerful complement to conventional quantitative investment models.

Within the domain of factor investing, machine learning applications can be broadly categorized into two directions:

1. **Factor Discovery** – By utilizing methods such as Genetic Programming (GP) and deep learning, researchers can automatically combine and derive new forms of "smart

factors," uncovering potential sources of excess returns [6]. The integration of alternative data—such as social media sentiment, search trends, and financial news—further enriches the scope of factor exploration [10].

2. Model Optimization – Through ensemble learning approaches (e.g., Random Forests [16], Gradient Boosting Trees [17], [18]) and deep architectures (e.g., Neural Networks, LSTM) [13], investors can construct predictive models with greater flexibility and precision, particularly useful in high-frequency trading, risk forecasting, and portfolio optimization.

Overall, the incorporation of machine learning into factor investing not only enhances the modeling of nonlinear relationships but also signifies a paradigm shift from theory-driven to data-driven approaches in investment research [1]. Future studies are expected to focus on reconciling model interpretability with predictive flexibility, thereby improving the stability, robustness, and practical value of investment strategies.

## 2 Methodology

### 2.1 Research Procedures

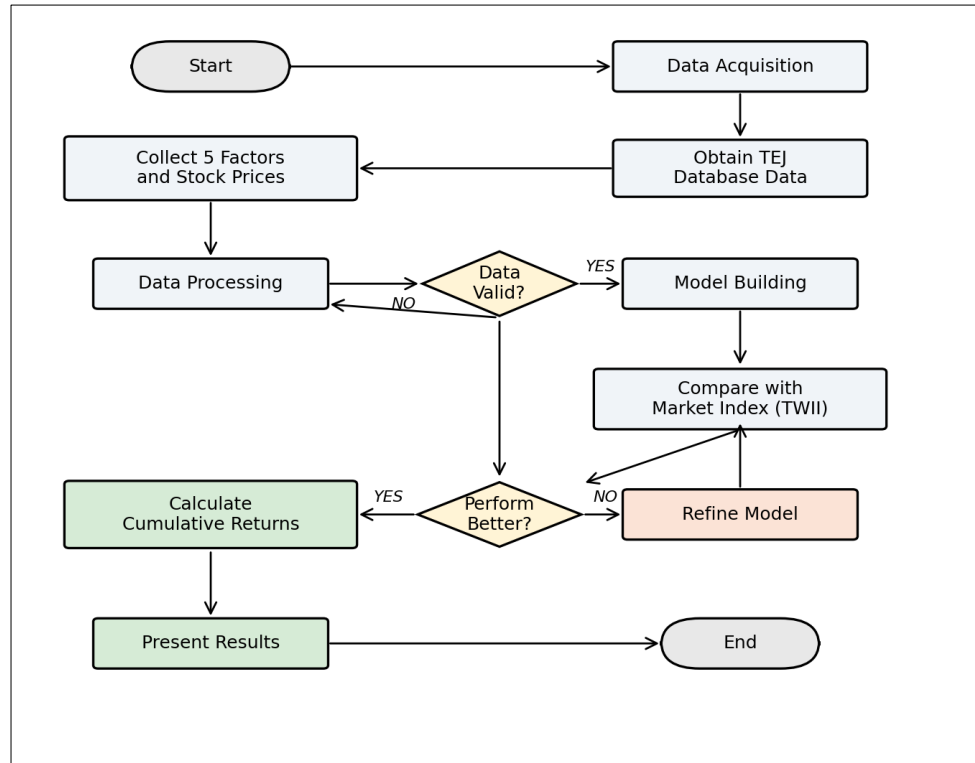


Figure1. Research Flowchart

### 2.2 Factor Selection

#### 2.2.1 One-Month Momentum (mom1m)

Jegadeesh and Titman [7] used past one-month stock returns as a basis for stock selection to predict future short-term performance (typically over several months up to one year). Within this framework, stocks are usually ranked according to their past returns and then divided into different groups (winners and losers) to examine the existence of momentum effects [7]. Stocks with higher returns in the past month tend to continue performing well in the short term ("winners keep winning"), whereas stocks with lower returns in the past month tend to continue underperforming ("losers keep losing") [7], [15].

The formula is expressed as:

$$x_{1,t} = mom1m = \ln \left( \frac{S_t}{S_{t-1}} \right) \quad (1)$$

### 2.2.2 Twelve-Month Momentum (mom12m)

Based on the cumulative returns over the past twelve months, stocks are classified into winner and loser portfolios, using past performance to predict future short-term return trends [8].

Jegadeesh [8] hypothesized that stocks with strong performance over the past twelve months (winner portfolio) tend to continue performing well in the short term, whereas stocks with weak performance over the past twelve months (loser portfolio) tend to continue underperforming.

Empirical findings indicate that the average returns of the winner portfolio during the holding period are significantly higher than those of the loser portfolio, suggesting that past twelve-month performance has predictive power for future returns [7], [8].

The formula is expressed as:

$$x_{2,t} = mom12m = \ln \left( \frac{S_{t-1}}{S_{t-12}} \right) \quad (2)$$

### 2.2.3 Change in Six-Month Momentum (chmom)

This factor measures how a stock's momentum has changed over the past six months, primarily capturing potential acceleration or deceleration trends within the momentum effect [11].

Gettleman and Marks [11] found that stocks in the high-chmom group typically earn significant excess returns in the short term, indicating that stocks with accelerating momentum have the potential to continue performing well. Conversely, stocks in the low-chmom group (with decelerating or reversing momentum) may underperform.

The formula is expressed as:

$$chmom_3 = Recent(mom6m) - Previous(mom6m) \quad (3)$$

### 2.2.4 Maximum Single-Period Return (maxret)

The maximum single-period return measures the largest price fluctuation of a stock within a given past period, reflecting the potential for extreme returns and short-term investment opportunities. This factor captures significant past price surges and can be used to predict potential trends in future short-term returns [14]. Studies indicate that stocks exhibiting high single-period returns in the past may continue to maintain strong momentum in the short term [7]. Particularly in high-volatility markets, such stocks



often attract momentum-following investors, reinforcing continued upward trends [14], [15].

The formula is expressed as:

$$maxret_{4,t} = max(R_{t-1}, R_{t-2}, \dots, R_{t-12}) \quad (4)$$

where  $R_{t-i}$  represents the monthly return of the stock in month  $t - i$ .

### 2.2.5 Return Volatility (retvol)

Return volatility measures the degree of variation in a stock's returns over a given period and serves as an indicator of investment risk [9]. Stocks with high volatility exhibit large price fluctuations, implying higher investment risk but also potentially higher return opportunities. Studies have found that stocks with low volatility and stable momentum tend to demonstrate more robust medium-term performance [9], whereas excessively volatile stocks are more prone to short-term reversals [14].

The formula is expressed as:

$$retvol_{5,t} = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (R_{t-i} - \bar{R})^2} \quad (5)$$

where  $R_{t-i}$  represents the stock return in month  $t - i$ ,  $\bar{R}$  is the average return over the period, and  $n$  is the number of observations.

## 2.3 Machine Learning Algorithm Selection

### 2.3.1 Ordinary Least Squares (OLS)

Ordinary Least Squares (OLS) is a method used to estimate unknown parameters in a linear regression model. It employs the principle of least squares to determine a set of parameters for the explanatory variables by minimizing the sum of squared residuals between the observed dependent variable and the values predicted by the model. OLS is widely used as a baseline benchmark in empirical asset pricing studies [1], [3].

OLS predicts a response variable using a set of predictor variables. In linear regression, the model is linear with respect to the parameters  $\beta$ :

$$y_{t+1} = \alpha + \beta x_t + \mu_t \quad (6)$$

where  $y_t$  is the dependent variable,  $x_t$  is the independent variable,  $\alpha$  and  $\beta$  are the parameters to be estimated via least squares (also called regression coefficients), and  $\mu_t$  is the random error term.

The fundamental principle of OLS linear regression is to minimize the Residual Sum of Squares (RSS), i.e., the sum of squared distances between observed points and the fitted line, to achieve the best fit:

$$RSS = \sum_{t=1}^T (y_t - \hat{y}_t)^2 = \sum_{t=1}^T (y_t - \hat{\alpha} - \hat{\beta} x_t)^2 \quad (7)$$

### 2.3.2 Random Forest (RF)

Random Forest, proposed by Breiman [16], is an ensemble learning algorithm that uses decision trees as base learners and can be applied to both classification and regression tasks. Its non-linear characteristics allow it to adapt effectively to various types of data and scenarios, making it widely adopted in asset pricing research [1].

Random Forest generates new training sets by repeatedly drawing (n) samples with replacement (bootstrap) from the original training dataset (N) [16]. For each sampled dataset, a decision tree is constructed. At each node of the tree, (d) features are randomly selected without replacement, and the best feature and split point are determined using criteria such as the Gini index, gain ratio, or information gain. This process is repeated (k) times, where (k) is the total number of decision trees in the forest.

Finally, predictions for the test samples are made using the trained Random Forest, and the final output is determined by a majority voting scheme [16].

The main characteristics of Random Forest are as follows [16]:

1. High parallelizability and computational efficiency: Since each decision tree is trained independently on a bootstrap sample, the training process can be parallelized, improving computational efficiency.
2. Inheritance of CART advantages: Random Forest is based on the CART algorithm and employs the Gini index to select optimal features and split points, enabling effective handling of complex nonlinear relationships.

3. Reduction of overfitting and improvement of generalization ability: By aggregating the predictions of multiple decision trees through the bagging mechanism, Random Forest reduces model variance, mitigates overfitting, and enhances predictive robustness on unseen data [16].

### 2.3.3 Neural Network (NNs)

Neural Networks are mathematical models inspired by the structure of the human brain, designed to handle prediction and classification tasks involving non-linear and high-dimensional data. The basic architecture consists of an input layer, one or more hidden layers, and an output layer. By using non-linear activation functions and learned weights, neural networks capture complex relationships between inputs and outputs. The inclusion of hidden layers endows neural networks with universal approximation capability [13], and increasing the number of layers (deep learning) further enhances model performance [13].

During training, optimization algorithms such as Stochastic Gradient Descent (SGD) and Adam [12], combined with regularization techniques like learning rate decay, early stopping, and batch normalization, improve both the generalization ability and training efficiency of the network [1], [12]. Neural networks have become a core tool in modern artificial intelligence and have been widely applied in asset pricing [1], [10].

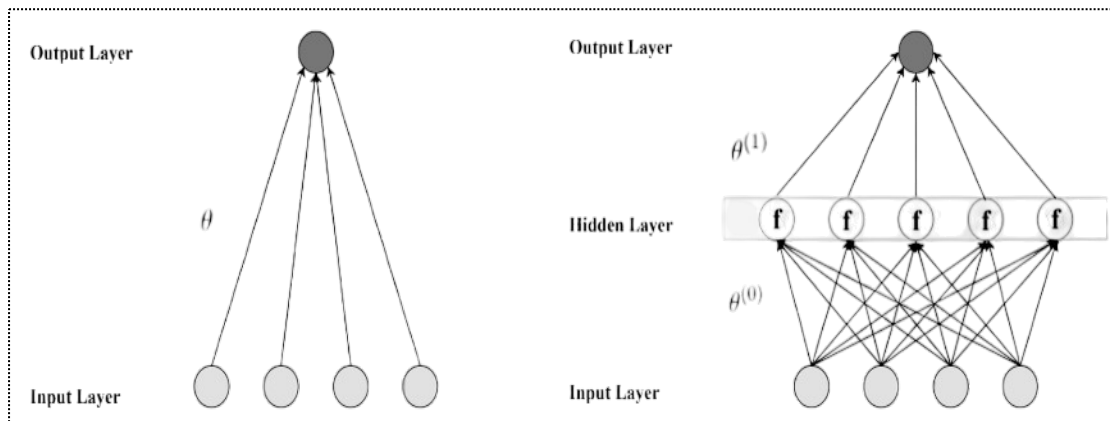


Figure 2. Schematic Diagram of Neural Network Principles

Figure 2 illustrates two simple examples of neural networks: one without a hidden layer (left) and one with a hidden layer (right). Gray circles represent the input layer, and black circles represent the output layer. Each arrow corresponds to a weight parameter. In networks with hidden layers, non-linear activation functions transform the inputs before passing them to the output layer.

The left diagram represents the simplest network model, which does not include a hidden layer. The input signals are weighted according to a five-dimensional parameter vector (including the intercept and weights for each predictor), and the output layer sums the weighted signals to produce the predicted result. This model is equivalent to a linear regression model.

The right diagram illustrates a more flexible model that includes a single hidden layer, consisting of five neurons. Each neuron in the hidden layer linearly combines all input signals and then applies a non-linear activation function  $f$  to transform the result, which is subsequently passed to the output layer for final prediction.

The hidden layer in the right diagram enables the neural network to capture more complex non-linear relationships. The output of each neuron in the hidden layer is computed as:

$$h_j = f(\sum_i w_{ij} x_i + b_j) \quad (8)$$

where  $x_i$  represents the input signals,  $w_{ij}$  are the weights,  $b_j$  is the bias term, and  $h_j$  is the hidden neuron output. The outputs from all hidden neurons are then weighted and summed in the output layer to generate the final prediction.

This design demonstrates the flexibility of neural network architecture: the inclusion of a hidden layer not only expands the model's capability to represent complex functions but also enhances performance in addressing non-linear problems [13]. Proper selection of the hidden layer's depth and the number of neurons allows a balance between model expressiveness and computational efficiency [1], [13].

### 2.3.4 Gradient Boosting Regression Trees (GBRT)

Gradient Boosting Regression Trees (GBRT) is a boosting model that combines multiple decision trees as base learners. Each decision tree is trained on the residuals of all previous trees, using the negative gradient of the loss function in the current model as an approximation of the residuals, and then fits a regression tree [1], [17]. The general algorithm is as follows:

Input: Training dataset  $T = \{(x_1, y_1), (x_2, y_2), \dots, (x_N, y_N)\}$ , loss function  $L(y, f(x))$

Output: Regression tree  $\hat{f}(x)$

1. Initialization: Estimate a constant value that minimizes the loss function, which corresponds to a tree with only a root node:

$$f_0(x) = \arg \min_c \sum_{i=1}^N L(y_i, c) \quad (9)$$

2. For  $m = 1, 2, \dots, M$ :

a. Compute the negative gradient of the loss function evaluated at the current model, which serves as an estimate of the residuals. For the squared loss function, this equals the residual; for general loss functions, it approximates the residuals. For  $i = 1, 2, \dots, N$ :

$$r_{mi} = - \left[ \frac{\partial L(y, f(x_i))}{\partial f(x_i)} \right]_{f(x)=f_{m-1}(x)} \quad (10)$$

b. Fit a regression tree to  $r_{mi}$ , obtaining leaf node regions  $R_{mj}$ ,  $j = 1, 2, \dots, J$ , to approximate the residuals.

c. Estimate the value of each leaf region via a line search to minimize the loss function. For  $j = 1, 2, \dots, J$ :

$$c_{mj} = \arg \min_c \sum_{x_i \in R_{mj}} L(y_i, f_{m-1}(x_i) + c) \quad (11)$$

d. Update the regression tree:

$$f_m(x) = f_{m-1}(x) + \sum_{j=1}^J c_{mj} I(x \in R_{mj}) \quad (12)$$

e. Output final model:

$$\hat{f}(x) = f_M(x) = \sum_{m=1}^M \sum_{j=1}^J c_{mj} I(x \in R_{mj}) \quad (13)$$

### 2.3.5 Light Gradient Boosting Machine (LightGBM)

LightGBM, proposed by Ke et al. [18], is a high-performance gradient boosting algorithm that serves as an improved version of Gradient Boosting Regression Trees (GBRT). Its main feature lies in the use of a histogram-based algorithm to accelerate the search for optimal feature split points [18]. Additionally, it adopts a leaf-wise growth strategy, meaning that at each iteration, the leaf node yielding the greatest reduction in loss is selected for splitting. Compared with the traditional level-wise approach, the leaf-wise method can reduce training error more efficiently; however, it requires limiting the maximum depth to prevent overfitting [18].

Similar to GBRT, LightGBM approximates the residuals using the negative gradient of the loss function (14) and updates the model iteratively as in (15), where  $c_{mj}$  represents the optimal value of the leaf node, determined through the histogram-based algorithm to identify the best split point [18].

$$f_m(x) = f_{m-1}(x) + \sum_{j=1}^J c_{mj} I(x \in R_{mj}) \quad (14)$$

$$r_{mi} = - \left[ \frac{\partial L(y_i, f(x_i))}{\partial f(x_i)} \right]_{f(x)=f_{m-1}(x)} \quad (15)$$

LightGBM offers several advantages, including faster computation speed, lower memory usage, and strong scalability for handling large-scale datasets [18].

### 2.3.6 eXtreme Gradient Boosting (XGBoost)

XGBoost is an improved and extended version of the Gradient Boosting Regression Tree (GBRT). Its core design philosophy is to maintain high model accuracy while enhancing training efficiency and preventing overfitting. It is widely used in academic research and industry, particularly excelling in large-scale data analysis and machine learning competitions [1], [17].

Unlike traditional GBRT, which only uses first-order gradients, XGBoost incorporates both first- and second-order gradient information in its algorithm design. This allows the model to more accurately approximate changes in the loss function during each iteration[17].

For the  $t$ -th iteration, the objective function of XGBoost can be expressed using a second-order Taylor expansion as in (16), and during each iteration, the update formula is given by (17), where  $\eta$  denotes the learning rate,  $w_j$  represents the weight of the  $k$ -th leaf node, and  $R_{mj}$  refers to the corresponding partition region [17].

$$Obj^{(t)} \approx \sum_{i=1}^N [g_i f_t(x_i) + \frac{1}{2} h_i f_t(x_i)^2] + \Omega(f_t) \quad (16)$$

$$f_m(x) = f_{m-1}(x) + \eta \sum_{j=1}^J w_j I(x \in R_{mj}) \quad (17)$$

The main advantages of XGBoost can be summarized as follows:

1. More accurate loss approximation: By incorporating second-order gradient information, XGBoost can select splitting points more efficiently.
2. Regularization design: Parameters  $\gamma$  and  $\lambda$  control the complexity of the trees, effectively preventing overfitting.
3. High-performance implementation: Supports parallel and distributed computation, enabling large-scale data processing.
4. Flexibility: Compatible with various loss functions and custom objective functions, suitable for regression, classification, and ranking tasks.
5. Missing value handling: Automatically learns the optimal direction for missing value splits, improving adaptability to incomplete data.

## 2.4 Logarithmic Return (Ln Return)

The rate of return used in this study is the logarithmic return, rather than the simple return, because the logarithmic return possesses the property of additivity. It is not affected by the base period or time interval, and the percentage change in both upward and downward movements is symmetric.

The concept is to take the natural logarithm of the ratio between the closing price of the next period and that of the current period, expressed as follows:

$$R_{t+h} = \ln \left( \frac{S_{t+h}}{S_t} \right) \quad (18)$$

where  $R_{t+h}$  represents the return over a holding period of  $h$  months,  $S_{t+h}$  denotes the stock closing price at time  $t + h$ , and  $S_t$  denotes the stock closing price at time  $t$ .

## 2.4 Information Coefficient (IC)

The Information Coefficient (IC) is an indicator used to measure the predictive power of a selected stock factor on future stock returns. It is widely applied in quantitative investment analysis [1], [6]. The IC value reflects the correlation between the factor values and future stock returns over a given period. The larger the absolute value of IC, the stronger the factor's predictive ability for subsequent returns. Therefore, IC serves as an important tool for evaluating the effectiveness of factors [1].

$$IC_t = \text{corr}(X_t, R_{t+1}) \quad (19)$$

Table 1. Summary of Correlation Judgment

| Correlation | Condition |
|-------------|-----------|
| Positive    | $IC > 0$  |
| Negative    | $IC < 0$  |
| None        | $IC = 0$  |

Table 2. Summary Statistics of Factors (%)

| Factor Abbreviation | IC Mean | IC Standard Deviation | Ln Return Mean | Ln Return Standard Deviation | IC Absolute Mean |
|---------------------|---------|-----------------------|----------------|------------------------------|------------------|
| mom1m               | 5.986   | 10.638                | 0.072          | 9.713                        | 5.986            |
| mom12m              | 12.748  | 7.156                 | 0.141          | 7.324                        | 12.748           |
| chmom               | 3.525   | 14.249                | 0.281          | 3.462                        | 3.525            |
| maxret              | -3.216  | 16.579                | 0.357          | 4.158                        | 3.216            |
| retvol              | -1.873  | 6.853                 | 0.095          | 6.158                        | 1.873            |

### 3 Data Sources and Processing

The data for this study are sourced from the Taiwan Economic Journal (TEJ) database, covering the period from January 1999 to February 2025 (monthly data over a total of 25 years). A total of 964 listed companies were considered, and monthly factor data were calculated for empirical analysis.

The historical IC values of the five factors were used as input data for six machine learning models. The independent variables  $X$  consist of the five factors, while the dependent variable  $Y$  corresponds to the IC values. The machine learning models are employed to predict the IC values of the factors for the next period. If the predicted IC value is positive, a “buy-high, sell-low” investment strategy is adopted (i.e., buying the companies in the highest group and selling those in the lowest group). Conversely, if the predicted IC value is negative, a “buy-low, sell-high” strategy is implemented (i.e., buying the companies in the lowest group and selling those in the highest group).



The market portfolios are divided into 10, 20, 50, 100, 192, and 964 groups, respectively, and their performance is compared with that of the Taiwan Weighted Index (TWII).

Table 3. Calculation Formulas for Factor Log Returns

| Factor      | Calculation Formula                      |
|-------------|------------------------------------------|
| All Factors | IFERROR(LN(Close!O2/Close!N2),"Missing") |

Table 4. Calculation Formulas for Next Month's Factor Returns

| Factor | Calculation Formula                                              |
|--------|------------------------------------------------------------------|
| mom1   | IFERROR(LN(Close!N2/Close!M2),"Missing")                         |
| mom12  | IFERROR(LN(Close!N2/Close!B2),"Missing")                         |
| chmom  | IFERROR(LN(Close!N2/Close!M2) - LN(Close!N2/Close!B2),"Missing") |
| maxret | IFERROR(MAX(C2:N2),"Missing")                                    |
| retvol | IFERROR(STDEV(C2:N2),"Missing")                                  |

Table 5. Filling Missing Values for Each Factor

| Factor | Calculation Formula                                                                                      |
|--------|----------------------------------------------------------------------------------------------------------|
| mom1   | IF(OR('mom1'!C2="Missing",NextMonthReturn!C2="Missing"),<br>MEDIAN('mom1'!C\$2:C\$965), 'mom1'!C2)       |
| mom12  | IF(OR('mom12'!C2="Missing",NextMonthReturn!C2="Missing"),<br>MEDIAN('mom12'!C\$2:C\$965), 'mom12'!C2)    |
| chmom  | IF(OR('chmom'!C2="Missing",NextMonthReturn!C2="Missing"),<br>MEDIAN('chmom'!C\$2:C\$965), 'chmom'!C2)    |
| maxret | IF(OR('maxret'!C2="Missing",NextMonthReturn!C2="Missing"),<br>MEDIAN('maxret'!C\$2:C\$965), 'maxret'!C2) |
| retvol | IF(OR('retvol'!C2="Missing",NextMonthReturn!C2="Missing"),<br>MEDIAN('retvol'!C\$2:C\$965), 'retvol'!C2) |

Table 6. Filling Next-Month Returns for Each Factor

| Factor | Calculation Formula                                                                                                     |
|--------|-------------------------------------------------------------------------------------------------------------------------|
| mom1   | IF(OR('mom1'!C2="Missing",NextMonthReturn!C2="Missing"),<br>MEDIAN('NextMonthReturn'!C\$2:C\$965),NextMonthReturn!C2)   |
| mom12  | IF(OR('mom12'!C2="Missing",NextMonthReturn!C2="Missing"),<br>MEDIAN('NextMonthReturn'!C\$2:C\$965),NextMonthReturn!C2)  |
| chmom  | IF(OR('chmom'!C2="Missing",NextMonthReturn!C2="Missing"),<br>MEDIAN('NextMonthReturn'!C\$2:C\$965),NextMonthReturn!C2)  |
| maxret | IF(OR('maxret'!C2="Missing",NextMonthReturn!C2="Missing"),<br>MEDIAN('NextMonthReturn'!C\$2:C\$965),NextMonthReturn!C2) |
| retvol | IF(OR('retvol'!C2="Missing",NextMonthReturn!C2="Missing"),<br>MEDIAN('NextMonthReturn'!C\$2:C\$965),NextMonthReturn!C2) |

Table 7. Information Coefficient Calculation for Each Factor

| Factor | Calculation Formula                                                      |
|--------|--------------------------------------------------------------------------|
| mom1   | CORREL(mom1_Imputed!C\$2:C\$965,<br>NextMonthReturn_Imputed!C\$2:C\$965) |
| mom12  | CORREL(mom1_Imputed!C\$2:C\$965,<br>NextMonthReturn_Imputed!C\$2:C\$965) |
| chmom  | CORREL(mom1_Imputed!C\$2:C\$965,<br>NextMonthReturn_Imputed!C\$2:C\$965) |
| maxret | CORREL(mom1_Imputed!C\$2:C\$965,<br>NextMonthReturn_Imputed!C\$2:C\$965) |
| retvol | CORREL(mom1_Imputed!C\$2:C\$965,<br>NextMonthReturn_Imputed!C\$2:C\$965) |

## 4 Results Presentation

### 4.1 Investment Strategy Construction Based on Predicted IC

An investment strategy is constructed based on the predicted IC values. If the IC value is positive, a “buy high, sell low” strategy is adopted; if the IC value is negative, a “buy low, sell high” strategy is applied. The factor with the larger absolute IC value is used for the investment strategy, as it has a stronger influence on the next period’s returns.

Table 8.

| Time                          | 2013/12      | 2014/01      | 2014/02      | 2014/03      | ..... | 2024/10      | 2024/11     | 2024/12      | 2025/01      |
|-------------------------------|--------------|--------------|--------------|--------------|-------|--------------|-------------|--------------|--------------|
| mom1                          | -0.039       | 0.039        | -0.025       | -0.003       | ..... | -0.065       | 0.063       | -0.009       | 0.014        |
| mom12                         | -0.248       | -0.275       | -0.085       | 0.023        | ..... | 0.051        | 0.034       | 0.158        | 0.243        |
| chmom                         | 0.118        | -0.090       | 0.023        | 0.005        | ..... | 0.073        | 0.011       | 0.048        | 0.184        |
| retvol                        | 0.011        | 0.027        | 0.015        | 0.020        | ..... | 0.010        | 0.001       | -0.023       | 0.020        |
| maxret                        | -0.001       | 0.007        | 0.017        | -0.011       | ..... | -0.022       | 0.026       | -0.037       | 0.002        |
| Choose largest absolute value | 2            | 2            | 2            | 2            | ..... | 3            | 1           | 2            | 2            |
|                               | <b>mom12</b> | <b>mom12</b> | <b>mom12</b> | <b>mom12</b> | ..... | <b>chmom</b> | <b>mom1</b> | <b>mom12</b> | <b>mom12</b> |
|                               | -0.248       | -0.275       | -0.085       | 0.023        | ..... | 0.073        | 0.063       | 0.158        | 0.243        |

### 4.2 Portfolio Returns with Different Quantile Divisions

This study constructs portfolio returns based on the ranking results of each month and compares the effects of different grouping methods on portfolio performance, in order to examine the variations in portfolio performance across different quantile divisions.

In addition, historical data of the Taiwan Weighted Index is downloaded and its logarithmic returns are calculated to further observe whether the constructed portfolios exhibit performance that outperforms the broader market.

Table 9.

| Equal Partition | Number of Firms | 2014/01 | 2014/02 | 2014/03 | 2014/04 | ..... | 2024/11 | 2024/12 | 2025/01 | 2025/02 |
|-----------------|-----------------|---------|---------|---------|---------|-------|---------|---------|---------|---------|
| 10              | 96              | 0.014   | -0.015  | 0.001   | -0.056  | ..... | 0.022   | 0.028   | -0.028  | -0.032  |
| 20              | 48              | -0.007  | 0.005   | 0.006   | -0.060  | ..... | 0.001   | 0.012   | -0.036  | -0.048  |
| 50              | 19              | -0.033  | 0.142   | 0.048   | -0.060  | ..... | -0.007  | 0.033   | -0.043  | -0.085  |
| 100             | 10              | -0.036  | 0.273   | 0.095   | 0.035   | ..... | 0.014   | 0.059   | -0.049  | -0.160  |
| 107             | 9               | -0.039  | 0.234   | 0.087   | 0.079   | ..... | 0.016   | 0.058   | -0.059  | -0.158  |
| 120             | 8               | -0.039  | 0.169   | 0.097   | 0.081   | ..... | -0.010  | 0.075   | -0.064  | -0.140  |
| 137             | 7               | -0.059  | 0.184   | 0.100   | 0.119   | ..... | 0.010   | 0.081   | -0.053  | -0.155  |
| 160             | 6               | -0.074  | 0.220   | 0.141   | 0.178   | ..... | 0.051   | 0.089   | -0.023  | -0.168  |
| 192             | 5               | 0.011   | 0.223   | 0.146   | 0.206   | ..... | 0.060   | 0.125   | -0.020  | -0.210  |
| 241             | 4               | 0.010   | 0.325   | 0.044   | 0.184   | ..... | 0.132   | 0.191   | 0.011   | -0.179  |
| 321             | 3               | 0.026   | 0.417   | 0.022   | 0.068   | ..... | 0.126   | 0.207   | 0.036   | -0.161  |
| 482             | 2               | 0.031   | -0.089  | 0.015   | 0.182   | ..... | 0.331   | 0.063   | 0.070   | -0.083  |
| 964             | 1               | 0.028   | -0.082  | -0.082  | 0.202   | ..... | 0.143   | 0.293   | -0.036  | -0.129  |
|                 | TW II           | -0.017  | 0.021   | 0.024   | -0.007  | ..... | -0.025  | 0.034   | 0.021   | -0.020  |

## 5 Results Analysis

### 5.1 Winning Percentage

This study constructs investment portfolios of different equal partitions using various machine learning models and examines their performance relative to the market index. If a portfolio's monthly return exceeds that of the market, it is recorded as 1; otherwise, it is recorded as 0.

Subsequently, the results for all months are summed and divided by 134 (the total number of months in the sample period) to calculate the winning percentage. A winning percentage greater than 0.5 indicates that the partitioning method demonstrates a better investment strategy performance under the corresponding model.

Table 10.

| Equal Partition    | Number of Firms | OLS          | RF           | LightGBM     | GBRT         | XGBoost      | NN1       | NN2          | NN3          | NN4          | NN5          | Real         |
|--------------------|-----------------|--------------|--------------|--------------|--------------|--------------|-----------|--------------|--------------|--------------|--------------|--------------|
| 10                 | 96              | 0.403        | <b>0.530</b> | 0.373        | 0.425        | 0.448        | 0.425     | <b>0.522</b> | <b>0.500</b> | <b>0.515</b> | 0.448        | <b>0.993</b> |
| 20                 | 48              | 0.440        | <b>0.545</b> | 0.433        | 0.433        | 0.485        | 0.433     | <b>0.560</b> | <b>0.522</b> | <b>0.515</b> | 0.493        | <b>0.955</b> |
| 50                 | 19              | 0.478        | <b>0.582</b> | <b>0.515</b> | 0.403        | 0.470        | 0.455     | <b>0.545</b> | <b>0.552</b> | <b>0.507</b> | 0.485        | <b>0.873</b> |
| 100                | 10              | <b>0.515</b> | <b>0.612</b> | <b>0.515</b> | 0.440        | <b>0.522</b> | 0.448     | <b>0.537</b> | 0.485        | <b>0.522</b> | <b>0.500</b> | <b>0.836</b> |
| 107                | 9               | <b>0.545</b> | <b>0.590</b> | 0.478        | 0.455        | <b>0.507</b> | 0.440     | <b>0.545</b> | 0.485        | <b>0.515</b> | <b>0.500</b> | <b>0.821</b> |
| 120                | 8               | <b>0.537</b> | <b>0.619</b> | <b>0.507</b> | 0.463        | 0.485        | 0.470     | <b>0.567</b> | <b>0.507</b> | <b>0.552</b> | <b>0.530</b> | <b>0.821</b> |
| 137                | 7               | <b>0.522</b> | <b>0.627</b> | <b>0.530</b> | 0.485        | <b>0.507</b> | 0.485     | <b>0.537</b> | <b>0.537</b> | <b>0.537</b> | 0.493        | <b>0.821</b> |
| 160                | 6               | <b>0.522</b> | <b>0.612</b> | <b>0.537</b> | 0.493        | <b>0.515</b> | 0.470     | <b>0.560</b> | <b>0.500</b> | <b>0.530</b> | 0.493        | <b>0.769</b> |
| 192                | 5               | <b>0.507</b> | <b>0.642</b> | <b>0.567</b> | 0.485        | <b>0.537</b> | 0.470     | <b>0.575</b> | <b>0.500</b> | <b>0.560</b> | <b>0.530</b> | <b>0.776</b> |
| 241                | 4               | 0.493        | <b>0.627</b> | <b>0.515</b> | <b>0.537</b> | <b>0.522</b> | 0.470     | <b>0.582</b> | <b>0.500</b> | <b>0.545</b> | <b>0.552</b> | <b>0.791</b> |
| 321                | 3               | <b>0.507</b> | <b>0.575</b> | <b>0.530</b> | <b>0.500</b> | 0.485        | 0.388     | <b>0.552</b> | 0.478        | <b>0.500</b> | <b>0.522</b> | <b>0.769</b> |
| 482                | 2               | 0.493        | <b>0.545</b> | 0.470        | 0.478        | <b>0.507</b> | 0.433     | <b>0.515</b> | 0.493        | <b>0.507</b> | <b>0.507</b> | <b>0.709</b> |
| 964                | 1               | 0.463        | <b>0.582</b> | 0.478        | 0.478        | <b>0.545</b> | 0.425     | <b>0.567</b> | 0.493        | <b>0.567</b> | <b>0.530</b> | <b>0.664</b> |
| Winning Percentage |                 | <b>54%</b>   | <b>100%</b>  | <b>62%</b>   | 15%          | <b>62%</b>   | 0%        | <b>100%</b>  | <b>62%</b>   | <b>100%</b>  | <b>77%</b>   | <b>100%</b>  |
| Ranking            |                 | <b>9</b>     | <b>1</b>     | <b>6</b>     | <b>10</b>    | <b>6</b>     | <b>11</b> | <b>1</b>     | <b>6</b>     | <b>1</b>     | <b>5</b>     | <b>1</b>     |

Through the application of various machine learning and traditional regression models to portfolio evaluation, the results in Table 10 indicate that the Random Forest (RF), two-layer neural network (NN2), and four-layer neural network (NN4) achieved a 100% winning percentage across all sample periods, demonstrating exceptional and stable investment performance.

Furthermore, the three-layer neural network (NN3), five-layer neural network (NN5), LightGBM, and XGBoost also achieved winning percentages exceeding 50%, suggesting that these models effectively capture market patterns and possess strong potential to outperform the overall market in the long run.

In contrast, the traditional Ordinary Least Squares (OLS) model, while not performing as well as the nonlinear models, still achieved a winning percentage above 50%, indicating that it retains explanatory power under certain conditions.

Overall, the findings of this study highlight the superiority of nonlinear machine learning models in portfolio construction and market prediction, validating the strategic insights and innovative value derived from comparing different modeling approaches.

## 5.2 Sharpe Ratio

Measuring the excess return generated per unit of risk is an important metric used to evaluate whether the returns obtained from an asset or portfolio are commensurate with the risks undertaken. In other words, this metric reflects “how much excess return can be earned for each unit of risk taken,” making it an indispensable benchmark in investment performance evaluation.

$$\text{Sharpe Ratio} = \frac{R_p - R_f}{\sigma_p} \quad (20)$$

$R_p$ : average return of the portfolio

$R_f$ : risk – free rate (e. g., Treasury bill rate)

$\sigma_p$ : standard deviation of the portfolio returns (risk)

Table 11.

| Sharpe Ratio Value | Interpretation                                                              |
|--------------------|-----------------------------------------------------------------------------|
| > 1                | Good (the return-to-risk ratio is reasonable)                               |
| > 2                | Excellent (risk-adjusted return is high)                                    |
| > 3                | Outstanding (rare)                                                          |
| < 1                | Poor (returns do not adequately compensate for risk)                        |
| < 0                | Very poor (risk is taken but returns are worse than a risk-free investment) |

Table 12 presents the Sharpe ratios of investment portfolios constructed under different machine learning models, using the market index as a benchmark for comparison. The results indicate that as the number of portfolio partitions increases, the performance of each model varies. Among them, Random Forest and gradient boosting models generally demonstrate superior performance, particularly in the top 10 and 20 partitions, where their Sharpe ratios are clearly higher than the market benchmark and approach the actual values.

In the top 10 partition, the actual Sharpe ratio is 1.706, while RF and XGBoost achieve 0.337 and 0.293, respectively. Although still below the actual performance, these values indicate that the models can effectively capture part of the market signals. Similarly, in the top 20 partition, the actual Sharpe ratio is 1.475, whereas RF and XGBoost reach 0.427 and 0.300, both significantly higher than the market benchmark of 0.174, highlighting the predictive potential of these models.

In contrast, the traditional OLS model and some neural network results are relatively unstable, with certain partitions even exhibiting negative values, indicating their limitations in capturing the nonlinear characteristics of the market. However, it is noteworthy that a few neural network models (NN2, NN3) still demonstrate Sharpe ratios above the market benchmark in some groups, suggesting that deep learning retains potential in specific contexts.

Overall, the empirical results indicate that machine learning models, particularly Random Forest, provide relatively stable and outstanding risk-adjusted returns in portfolio construction. This not only underscores the significance of nonlinear models

in financial market forecasting but also validates the practical value of machine learning methods in investment decision-making.

Table 12.

| Equal Partition | Number of Firms | OLS          | RF           | LightGBM | GBRT         | XGBoost      | NN1    | NN2          | NN3          | NN4          | NN5          | Real         |
|-----------------|-----------------|--------------|--------------|----------|--------------|--------------|--------|--------------|--------------|--------------|--------------|--------------|
| 10              | 96              | 0.023        | <b>0.337</b> | -0.050   | -0.016       | <b>0.293</b> | 0.034  | <b>0.386</b> | <b>0.255</b> | <b>0.326</b> | <b>0.220</b> | <b>1.706</b> |
| 20              | 48              | 0.050        | <b>0.427</b> | 0.035    | 0.051        | <b>0.300</b> | 0.062  | <b>0.427</b> | <b>0.262</b> | <b>0.343</b> | <b>0.238</b> | <b>1.475</b> |
| 50              | 19              | 0.144        | <b>0.429</b> | 0.045    | 0.052        | <b>0.294</b> | 0.050  | <b>0.255</b> | <b>0.278</b> | <b>0.198</b> | 0.134        | <b>1.077</b> |
| 100             | 10              | 0.145        | <b>0.397</b> | 0.010    | 0.065        | <b>0.298</b> | 0.035  | 0.125        | <b>0.217</b> | 0.128        | 0.064        | <b>0.867</b> |
| 107             | 9               | 0.110        | <b>0.388</b> | 0.016    | 0.080        | <b>0.299</b> | 0.017  | 0.150        | <b>0.233</b> | 0.141        | 0.117        | <b>0.813</b> |
| 120             | 8               | 0.141        | <b>0.340</b> | 0.029    | 0.088        | <b>0.307</b> | -0.004 | 0.139        | <b>0.227</b> | 0.128        | 0.107        | <b>0.813</b> |
| 137             | 7               | 0.148        | <b>0.328</b> | 0.019    | 0.021        | <b>0.310</b> | -0.013 | 0.131        | <b>0.208</b> | 0.129        | 0.090        | <b>0.843</b> |
| 160             | 6               | <b>0.178</b> | <b>0.306</b> | 0.015    | 0.077        | <b>0.309</b> | -0.037 | 0.124        | <b>0.199</b> | 0.097        | 0.049        | <b>0.778</b> |
| 192             | 5               | 0.134        | <b>0.279</b> | 0.086    | 0.070        | <b>0.294</b> | -0.071 | 0.163        | <b>0.202</b> | 0.060        | 0.033        | <b>0.777</b> |
| 241             | 4               | 0.072        | <b>0.255</b> | 0.080    | 0.105        | <b>0.299</b> | -0.083 | 0.124        | 0.154        | 0.090        | 0.076        | <b>0.754</b> |
| 321             | 3               | 0.064        | <b>0.180</b> | 0.021    | <b>0.183</b> | <b>0.284</b> | -0.169 | 0.081        | 0.140        | 0.040        | 0.043        | <b>0.695</b> |
| 482             | 2               | 0.037        | 0.127        | 0.012    | 0.120        | <b>0.288</b> | -0.104 | 0.049        | 0.119        | 0.066        | 0.036        | <b>0.530</b> |
| 964             | 1               | -0.023       | <b>0.187</b> | 0.075    | 0.142        | <b>0.306</b> | -0.092 | 0.098        | 0.113        | 0.106        | 0.034        | <b>0.439</b> |
|                 | TWII            | 0.174        | 0.174        | 0.174    | 0.174        | 0.174        | 0.174  | 0.174        | 0.174        | 0.174        | 0.174        | 0.174        |

### 5.3 Out-of-sample $R^2$

Out-of-sample  $R^2$  is an indicator used to evaluate the performance of a predictive model on data that was not involved in training. It primarily assesses the model's "generalization ability," or its predictive accuracy on truly unseen data. A higher value indicates a more reliable model that can provide forecasts more informative than simply using the mean.

$$R_{\text{out-of-sample}}^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2} \quad (21)$$

$R_{\text{out}}^2 > 0$ : indicates that the model predicts better than using the mean, demonstrating predictive power.

$R_{\text{out}}^2 < 0$ : indicates that the model performs worse than simply using the mean.

Table 13 presents the  $R^2$  performance of each model in portfolio prediction. The results indicate significant differences among models in explaining market variance. Overall, Random Forest performs the best, demonstrating substantial explanatory power and highlighting its advantage in capturing the nonlinear structures of the market.

In comparison, other ensemble methods such as LightGBM and XGBoost exhibit some explanatory ability but are not as strong as Random Forest. The traditional OLS model shows very low or even negative values, indicating its inability to effectively capture market characteristics. These findings underscore the importance of nonlinear and ensemble methods in financial market forecasting.

Table 13.

| Model | OLS    | RF    | LightGBM | GBRT  | XGBoost | NN1    | NN2   | NN3   | NN4   | NN5   |
|-------|--------|-------|----------|-------|---------|--------|-------|-------|-------|-------|
| $R^2$ | -3.742 | 0.763 | 0.464    | 0.325 | 0.364   | -0.373 | 0.419 | 0.739 | 0.621 | 0.525 |

## 6 Research Results

### 6.1 Cumulative Returns of Models (as of February 2025)

Table 14.

| Equal Partition | Number of Firms | OLS    | RF    | LightGBM | GBRT   | XGBoost | NN1    | NN2   | NN3   | NN4   | NN5   | Real   |
|-----------------|-----------------|--------|-------|----------|--------|---------|--------|-------|-------|-------|-------|--------|
| 10              | 96              | 0.108  | 1.641 | -0.133   | -0.056 | -0.339  | 0.135  | 1.663 | 1.228 | 1.437 | 1.031 | 6.063  |
| 20              | 48              | 0.277  | 2.437 | 0.136    | 0.216  | -0.276  | 0.338  | 2.201 | 1.577 | 1.786 | 1.337 | 7.148  |
| 50              | 19              | 1.154  | 3.369 | 0.283    | 0.344  | -0.636  | 0.277  | 2.021 | 2.485 | 1.585 | 1.034 | 8.325  |
| 100             | 10              | 1.481  | 4.121 | 0.088    | 0.548  | -0.117  | 0.098  | 1.370 | 2.478 | 1.384 | 0.596 | 9.636  |
| 107             | 9               | 1.155  | 4.142 | 0.142    | 0.727  | -0.206  | -0.025 | 1.686 | 2.751 | 1.626 | 1.168 | 9.817  |
| 120             | 8               | 1.581  | 4.050 | 0.281    | 0.893  | -0.212  | -0.303 | 1.713 | 2.855 | 1.668 | 1.202 | 10.094 |
| 137             | 7               | 1.824  | 4.348 | 0.217    | 0.231  | -0.219  | -0.426 | 1.802 | 2.905 | 1.844 | 1.104 | 11.066 |
| 160             | 6               | 2.399  | 4.411 | 0.176    | 0.916  | -0.610  | -0.845 | 1.678 | 2.941 | 1.446 | 0.639 | 10.953 |
| 192             | 5               | 2.116  | 4.282 | 1.166    | 0.948  | -0.385  | -1.532 | 2.348 | 3.120 | 0.969 | 0.478 | 11.477 |
| 241             | 4               | 1.288  | 4.284 | 1.282    | 1.458  | -0.344  | -1.954 | 2.022 | 2.696 | 1.578 | 1.279 | 11.772 |
| 321             | 3               | 1.365  | 3.522 | 0.400    | 2.263  | 0.501   | -3.595 | 1.512 | 2.867 | 0.803 | 0.820 | 12.264 |
| 482             | 2               | 1.034  | 3.266 | 0.284    | 2.103  | 2.522   | -2.941 | 1.047 | 3.090 | 1.704 | 0.910 | 12.273 |
| 964             | 1               | -0.771 | 6.071 | 2.357    | 3.396  | 2.822   | -2.972 | 2.557 | 4.495 | 3.468 | 1.131 | 14.833 |
|                 | TW II           | 0.985  | 0.985 | 0.985    | 0.985  | 0.985   | 0.985  | 0.985 | 0.985 | 0.985 | 0.985 | 0.985  |

From the results in Table 14, Random Forest demonstrates the most outstanding performance across all portfolio partitions, particularly in the lower-ranked groups, where its cumulative returns even exceed the actual market returns. In comparison, neural network models show overall good performance; except for NN1, most neural networks achieve higher cumulative returns than other machine learning models across different portfolio partitions, indicating a certain level of stability and predictive ability under varying portfolio sizes.

Examining the cumulative returns of gradient boosting models (GBRT, LightGBM, XGBoost) across different portfolio partitions, XGBoost exhibits negative returns in many partitions, reflecting higher sensitivity to risk. However, as the number of partitions increases, its final cumulative returns reach a relatively high level, demonstrating potential and competitiveness for long-term portfolio performance.



Meanwhile, the traditional OLS model shows relatively stable cumulative returns but ends with negative final cumulative returns, underperforming Random Forest and some neural network models.

In summary, the models can be ranked by performance from highest to lowest as follows: **Random Forest > Neural Networks > Gradient Boosting Models > Ordinary Least Squares.**

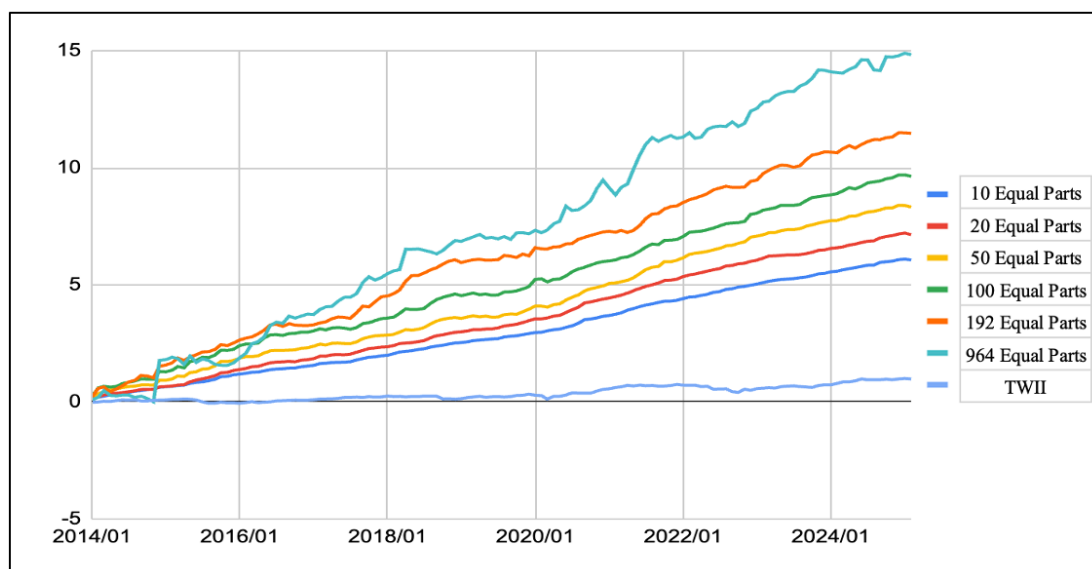


Figure 3. Actual Cumulative Returns

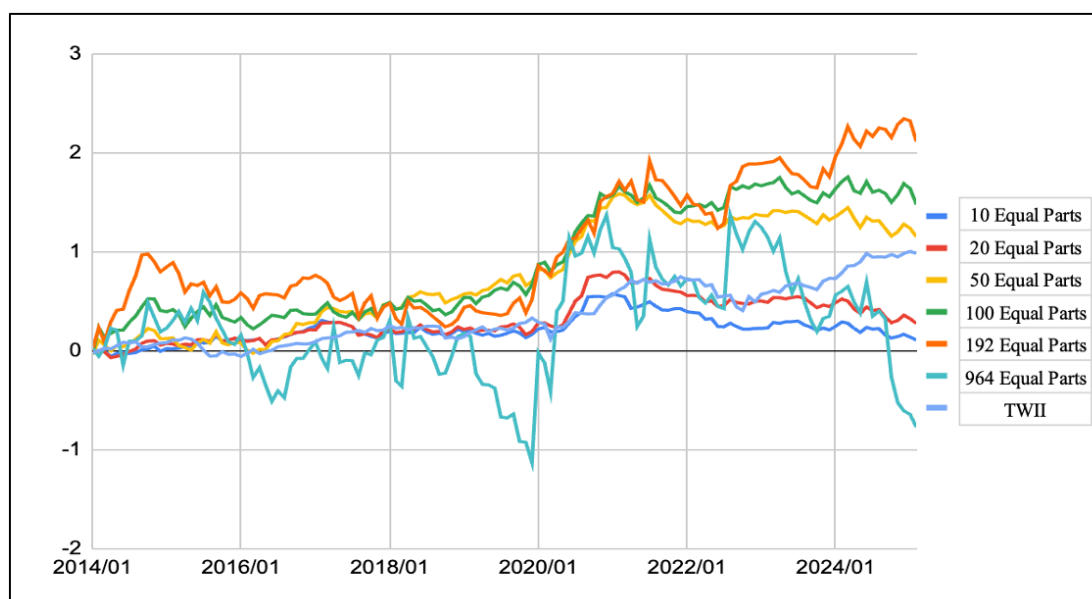


Figure 4. OLS Cumulative Returns

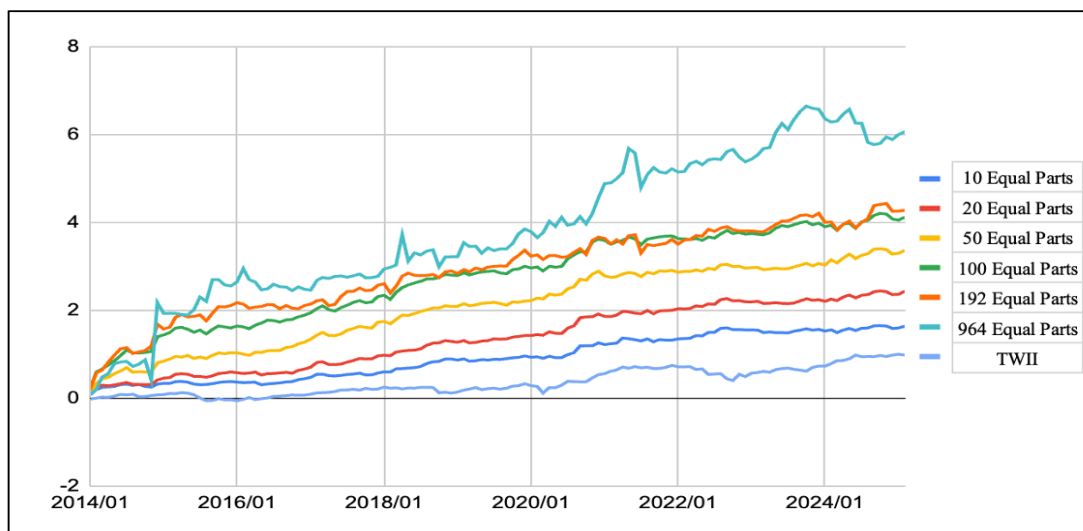


Figure 5. RF Cumulative Returns

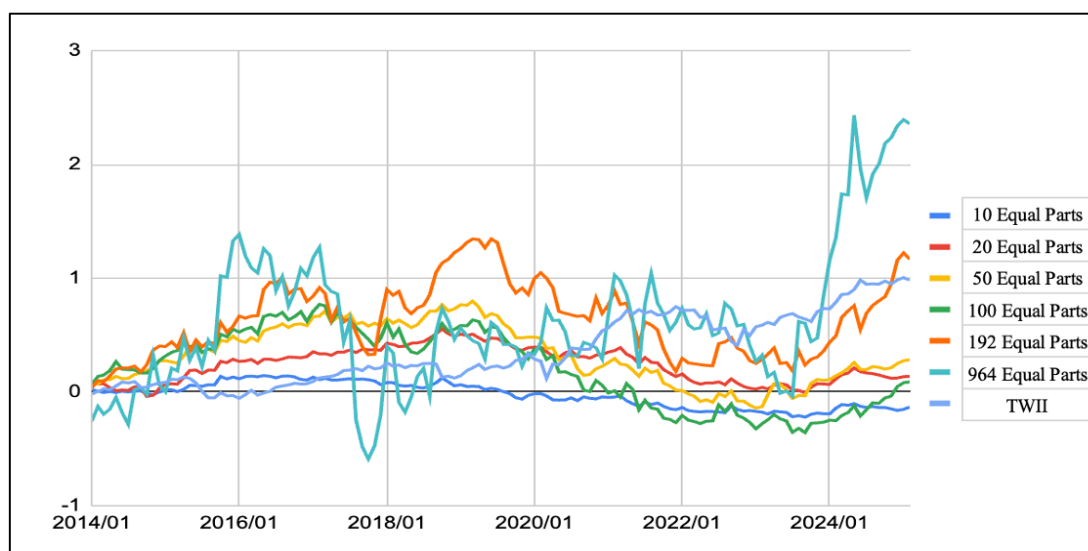


Figure 6. LightGBM Cumulative Returns

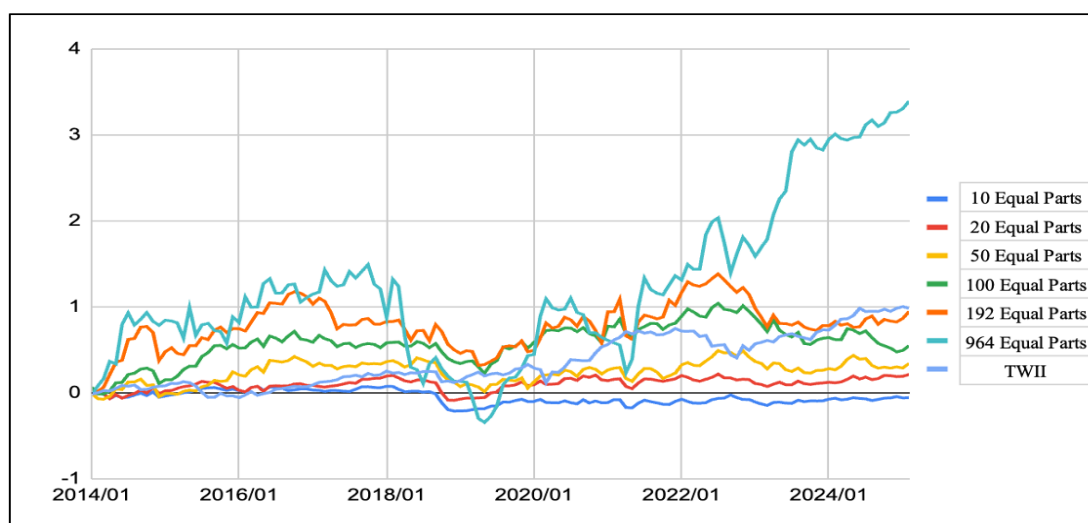


Figure 7. GBRT Cumulative Returns

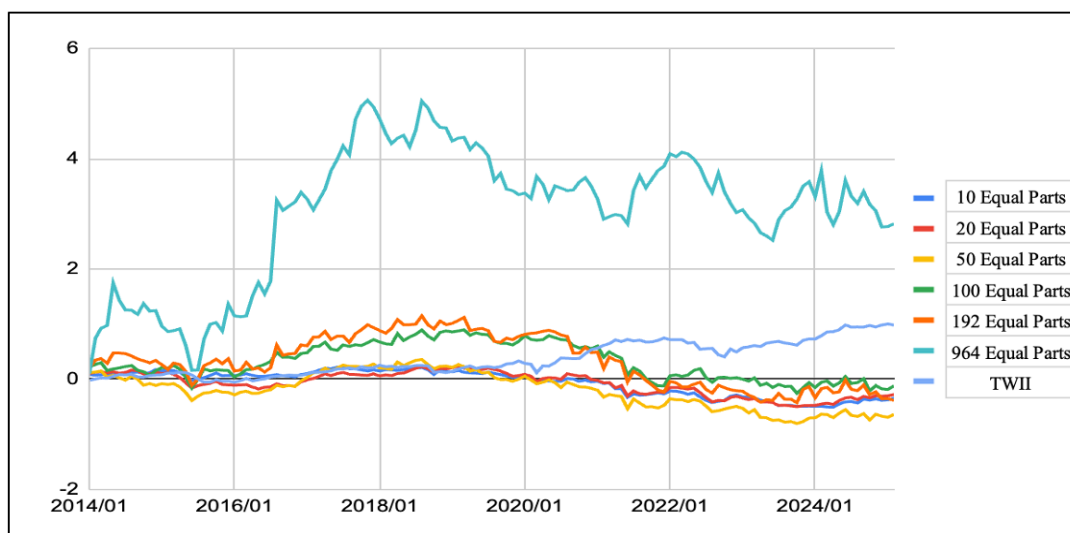


Figure 8. XGBoost Cumulative Returns

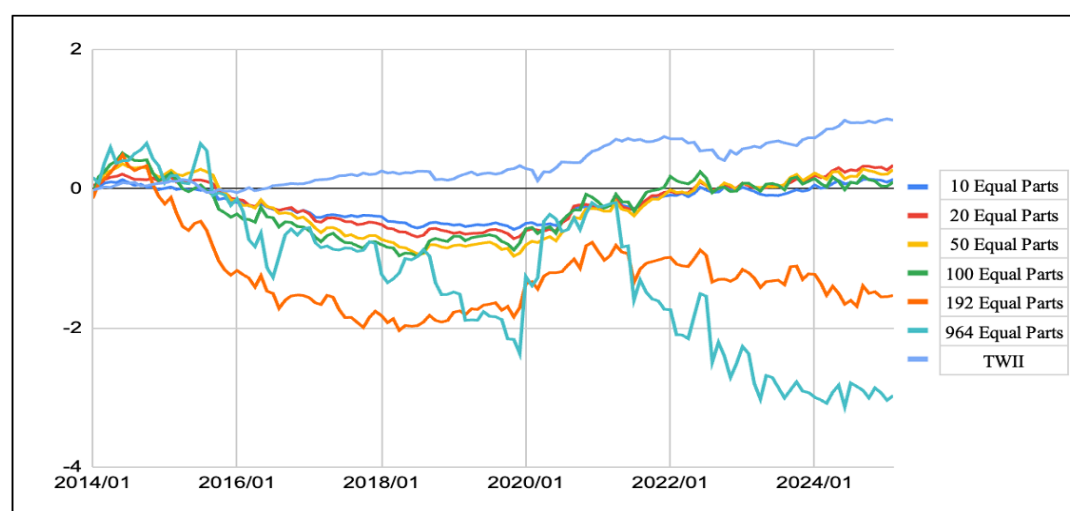


Figure 9. NN1 Cumulative Returns

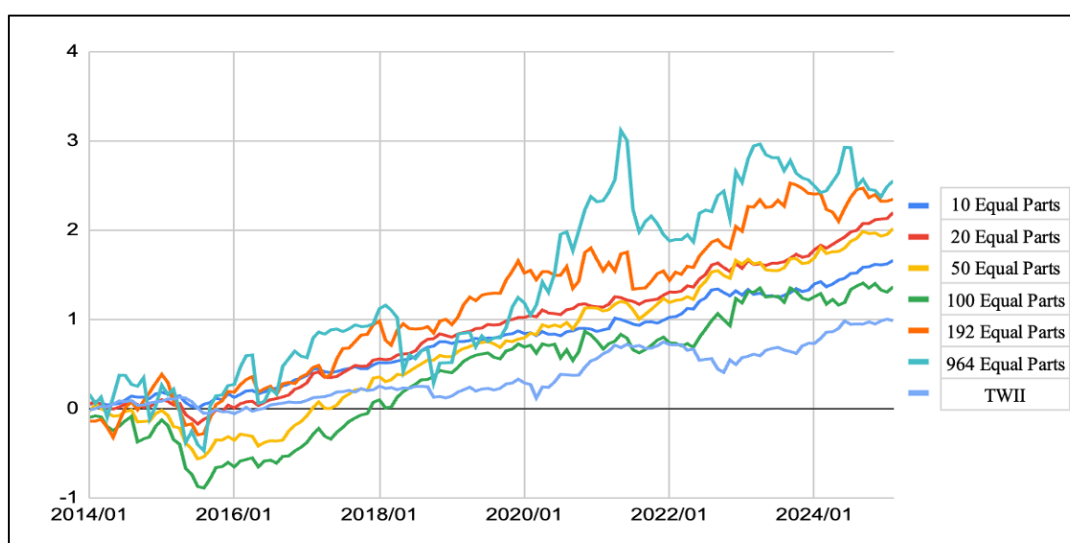


Figure 10. NN2 Cumulative Returns

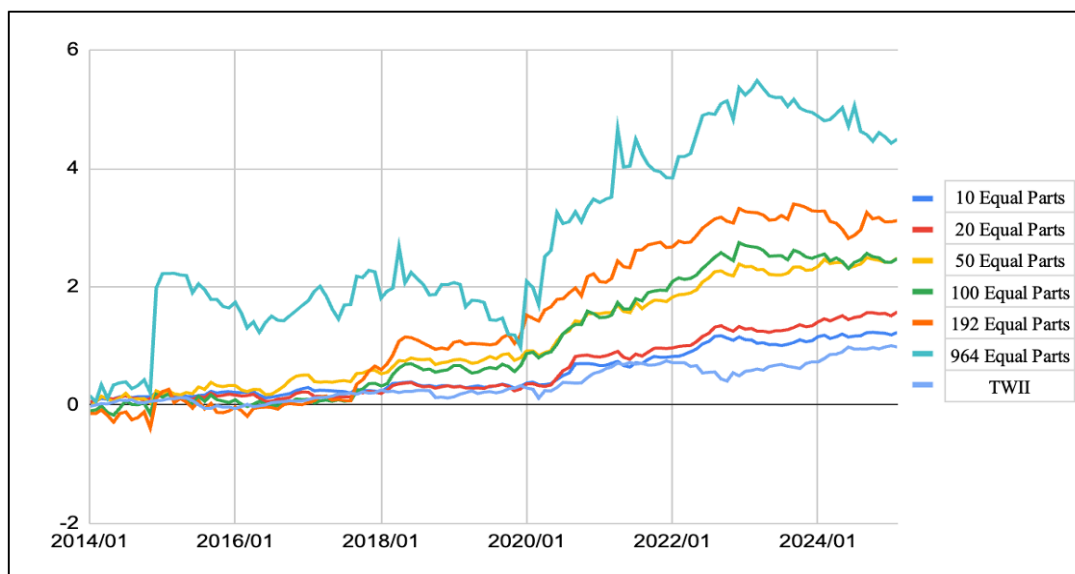


Figure 11. NN3 Cumulative Returns

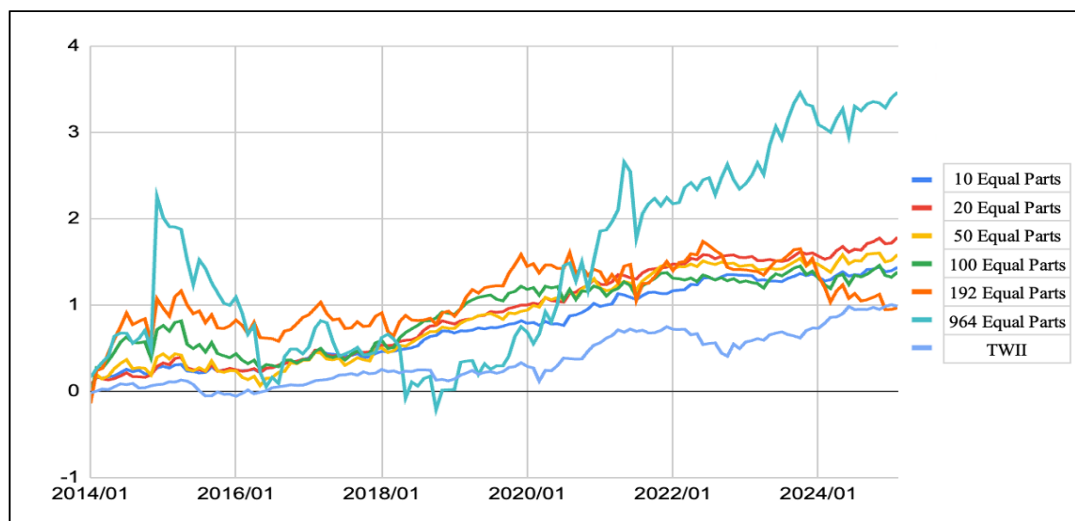


Figure 12. NN4 Cumulative Returns

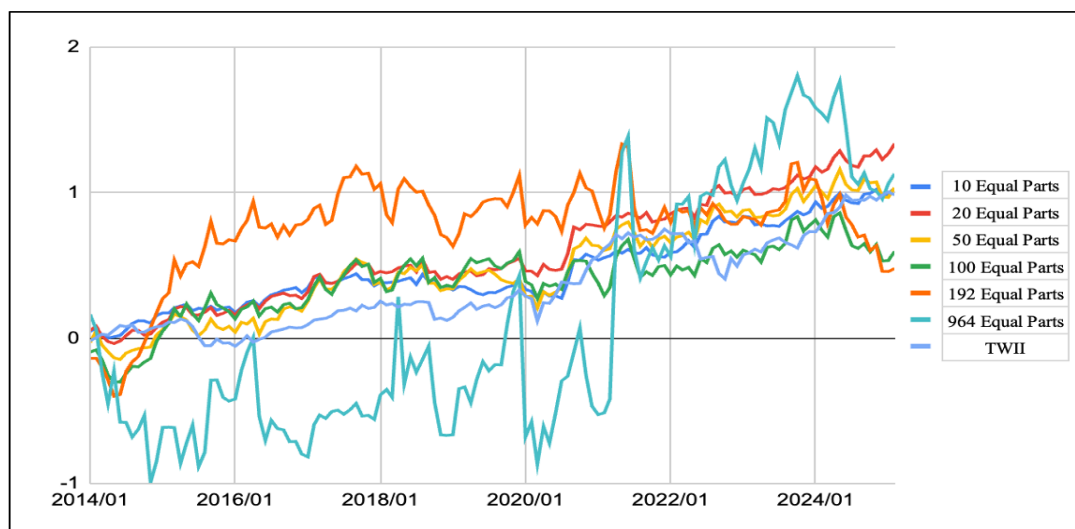


Figure 13. NN5 Cumulative Returns

## 6.2 Cumulative Returns by Equal Partition (as of February 2025)

Table 15.

| Model    | 10 Equal Parts | 20 Equal Parts | 50 Equal Parts | 100 Equal Parts | 192 Equal Parts | 964 Equal Parts |
|----------|----------------|----------------|----------------|-----------------|-----------------|-----------------|
| OLS      | 0.108          | 0.277          | 1.154          | 1.481           | 2.116           | -0.771          |
| RF       | 1.641          | 2.437          | 3.369          | 4.121           | 4.282           | 6.071           |
| GBRT     | -0.056         | 0.216          | 0.344          | 0.548           | 0.948           | 3.396           |
| XGBoost  | -0.339         | -0.276         | -0.636         | -0.117          | -0.385          | 2.822           |
| LightGBM | -0.133         | 0.136          | 0.283          | 0.088           | 1.166           | 2.357           |
| NN1      | 0.135          | 0.338          | 0.277          | 0.098           | -1.532          | -2.972          |
| NN2      | 1.663          | 2.201          | 2.021          | 1.370           | 2.348           | 2.557           |
| NN3      | 1.228          | 1.577          | 2.485          | 2.478           | 3.120           | 4.495           |
| NN4      | 1.437          | 1.786          | 1.585          | 1.384           | 0.969           | 3.468           |
| NN5      | 0.596          | 1.337          | 1.034          | 0.596           | 0.478           | 1.131           |
| Real     | 6.063          | 7.148          | 8.325          | 9.636           | 11.477          | 14.833          |
| TW II    | 0.985          | 0.985          | 0.985          | 0.985           | 0.985           | 0.985           |

Based on the results in Table 15, this study selected 10, 20, 50, 100, 192, and 964 partitions to analyze portfolio performance. The empirical results indicate that Random Forest maintained high robustness across all partition levels, consistently outperforming the market index and demonstrating strong adaptability under nonlinear structures. Neural networks exhibited flexible predictive power, effectively capturing market characteristics across different grouping scales, highlighting their advantage in complex data environments.

On the other hand, gradient boosting models (GBRT, XGBoost, LightGBM) showed limited performance in low-partition scenarios, but their performance gradually improved as the partitions became finer, particularly exhibiting notable excess returns at higher partition levels, reflecting their suitability for handling large-scale and high-dimensional market information. As for the traditional OLS model, although it achieved positive outcomes in certain scenarios, its overall performance remained inferior to machine learning methods, emphasizing its limitations in highly complex and nonlinear financial market structures.

In summary, the results in Table 15 further confirm that nonlinear machine learning models outperform linear approaches and validate the performance differences across various partition strategies, providing new empirical evidence for portfolio grouping design and investment strategy development.

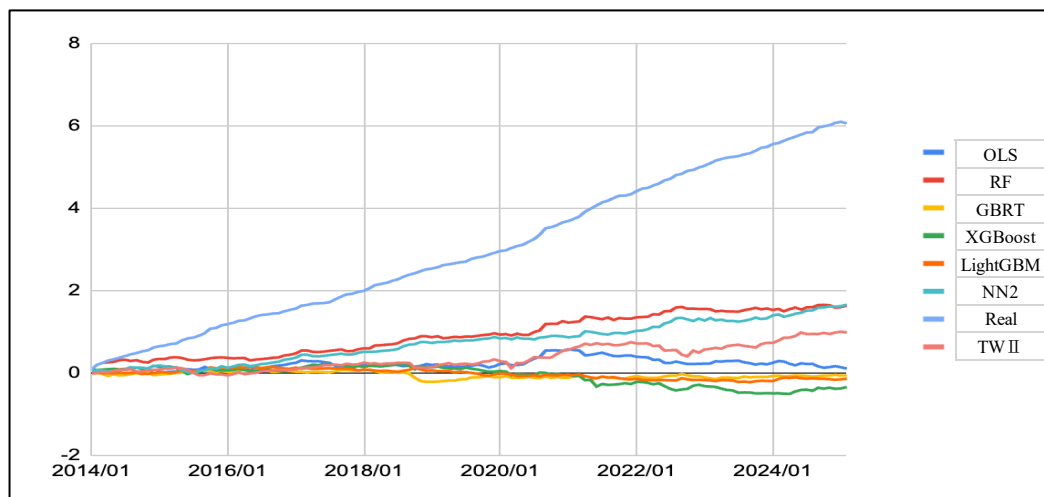


Figure 14. Cumulative Returns for 10 Equal Partitions

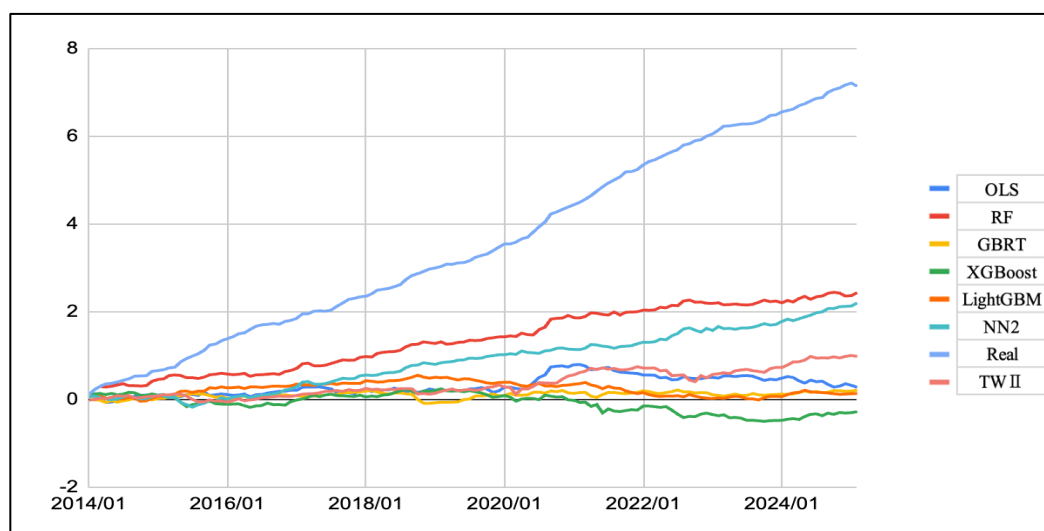


Figure 15. Cumulative Returns for 20 Equal Partitions

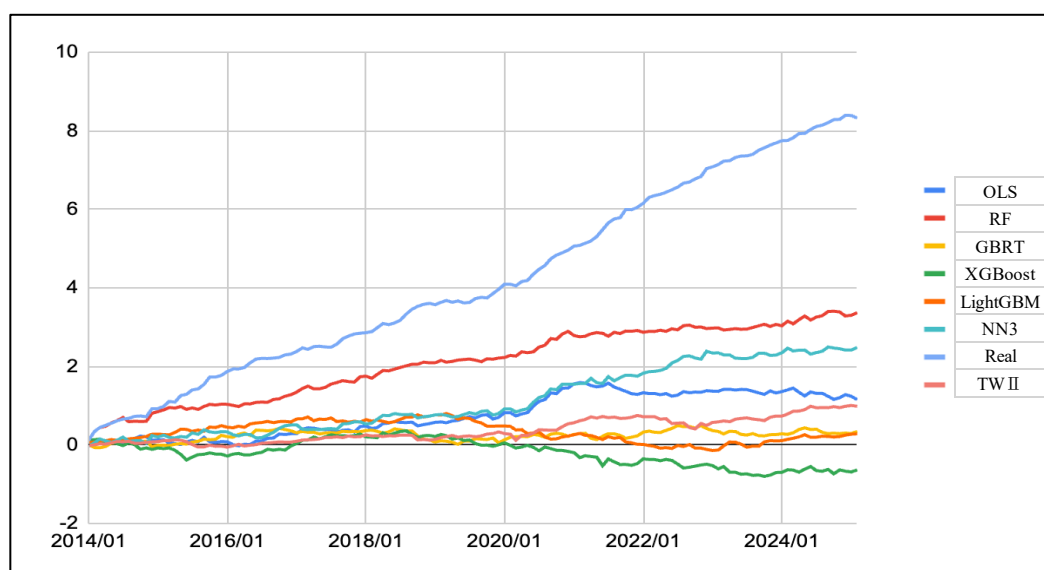


Figure 16. Cumulative Returns for 50 Equal Partitions

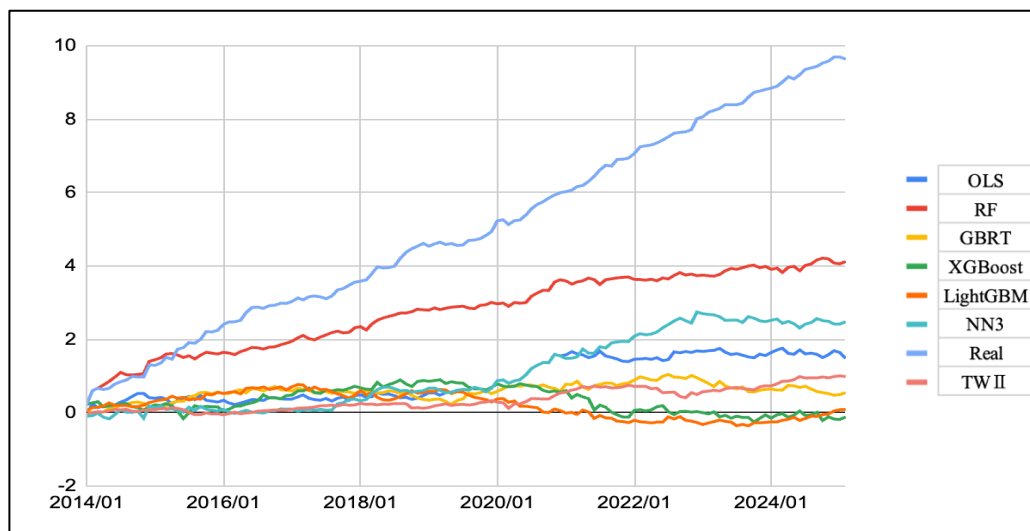


Figure 17. Cumulative Returns for 100 Equal Partitions

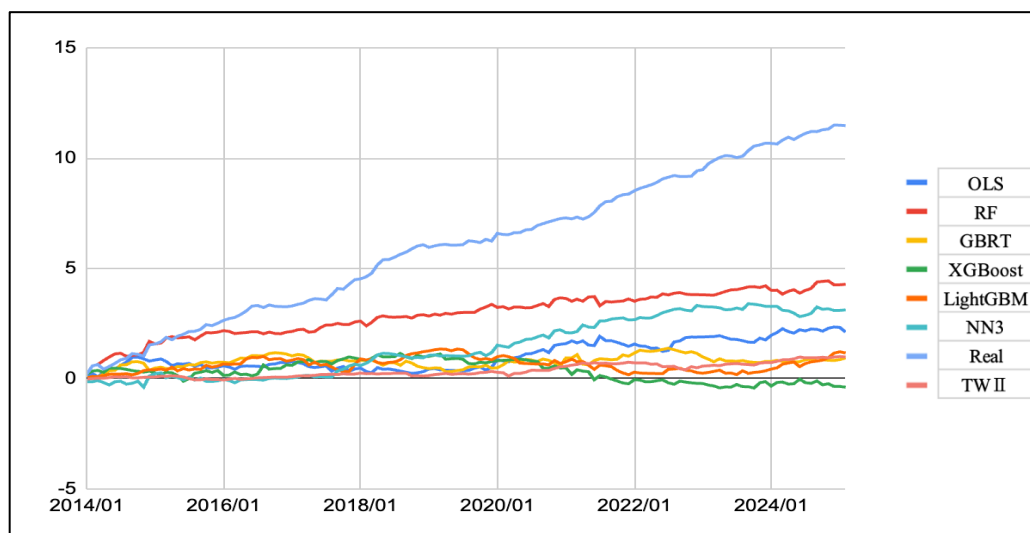


Figure 18. Cumulative Returns for 192 Equal Partitions

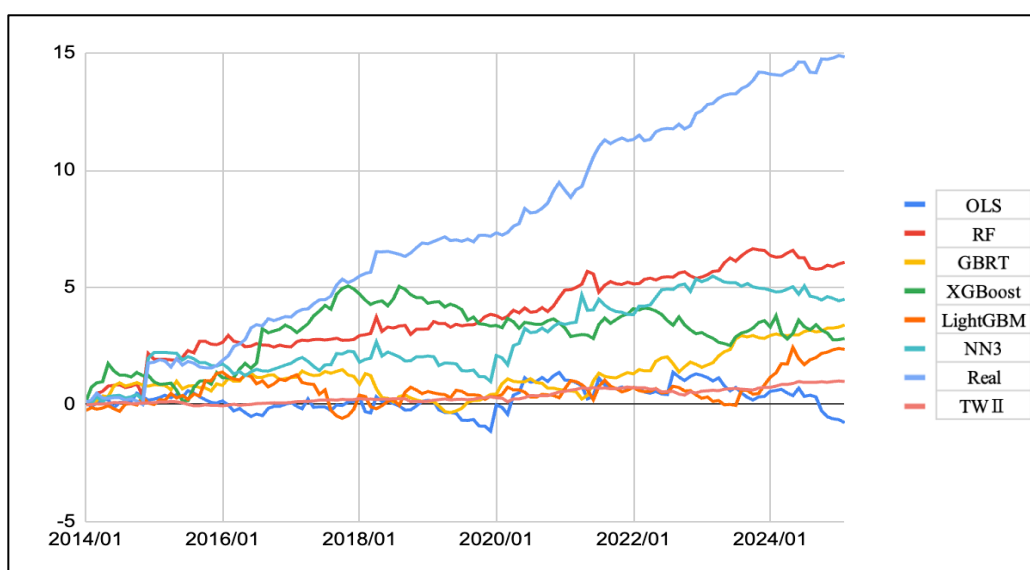


Figure 19. Cumulative Returns for 964 Equal Partitions

## **6.3 Stock Selection**

### **6.3.1 Stock Selection Strategy**

In this study, stock selection constitutes the core step in portfolio construction, aiming to identify stocks with investment potential through selected factors and machine learning models. Considering that different factors capture distinct market characteristics and risk preferences, this research conducts a comprehensive analysis based on five factors: short-term momentum, long-term momentum, momentum change, maximum single-period return, and return volatility. This approach is designed to balance return enhancement and risk management.

The stock selection process begins by ranking all stocks in the market according to their factor values. For each factor, stocks are divided into different quantile-based groups, from which those with superior performance are selected. Subsequently, six predictive models—linear regression, random forest, neural networks, gradient boosting regression trees, LightGBM, and XGBoost—are employed to model the relationship between the selected factors and future stock returns. These models generate expected returns for each stock.

Finally, based on the model prediction results, this study selects stocks from the highest and lowest quantile groups and identifies the corresponding company names as candidates for subsequent investment strategies. Depending on the strategy design, investors may choose to go long on high-score portfolios or low-score portfolios and adopt either “buy-low, sell-high” or “buy-high, sell-low” trading approaches. This methodology reflects both factor rankings and model-based predictions, providing a systematic and implementable stock selection framework that facilitates the evaluation of portfolio performance under different investment strategies.



### 6.3.2 Stock Selection Based on Model Predictions

Table 16. Real IC Selection Results

| Date            | 2014/01 | 2014/02 | 2014/03 | 2014/04 | ..... | 2024/11 | 2024/12 | 2025/01 | 2025/02 |
|-----------------|---------|---------|---------|---------|-------|---------|---------|---------|---------|
| Bottom-Quantile | 1101    | 2484    | 1233    | 2524    | ..... | 2611    | 1233    | 2379    | 2379    |
| Stock           | 台泥      | 希華      | 天仁      | 京城      | ..... | 志信      | 天仁      | 瑞昱      | 瑞昱      |
| Top-Quantile    | 9951    | 3219    | 2328    | 2489    | ..... | 2489    | 2328    | 1805    | 1440    |
| Stock           | 皇田      | 倚強科     | 廣宇      | 瑞軒      | ..... | 瑞軒      | 廣宇      | 寶徠      | 南紡      |

Table 17. OLS Selection Results

| Date            | 2014/01 | 2014/02 | 2014/03 | 2014/04 | ..... | 2024/11 | 2024/12 | 2025/01 | 2025/02 |
|-----------------|---------|---------|---------|---------|-------|---------|---------|---------|---------|
| Bottom-Quantile | 2524    | 1233    | 2524    | 1805    | ..... | 2538    | 3033    | 2385    | 3024    |
| Stock           | 京城      | 天仁      | 京城      | 寶徠      | ..... | 基泰      | 威健      | 群光      | 憶聲      |
| Top-Quantile    | 2489    | 2328    | 2436    | 2328    | ..... | 2436    | 2524    | 2524    | 2524    |
| Stock           | 瑞軒      | 廣宇      | 偉詮電     | 廣宇      | ..... | 偉詮電     | 京城      | 京城      | 京城      |

Table 18. RF Selection Results

| Date            | 2014/01 | 2014/02 | 2014/03 | 2014/04 | ..... | 2024/11 | 2024/12 | 2025/01 | 2025/02 |
|-----------------|---------|---------|---------|---------|-------|---------|---------|---------|---------|
| Bottom-Quantile | 1314    | 3024    | 3024    | 5521    | ..... | 2535    | 3033    | 5201    | 2535    |
| Stock           | 中石化     | 憶聲      | 憶聲      | 工信      | ..... | 達欣工     | 威健      | 凱衛      | 達欣工     |
| Top-Quantile    | 1451    | 2524    | 2524    | 2328    | ..... | 5007    | 2524    | 2524    | 5007    |
| Stock           | 年興      | 京城      | 京城      | 廣宇      | ..... | 三星      | 京城      | 京城      | 三星      |

Table 19. GBRT Selection Results

| Date            | 2014/01 | 2014/02 | 2014/03 | 2014/04 | ..... | 2024/11 | 2024/12 | 2025/01 | 2025/02 |
|-----------------|---------|---------|---------|---------|-------|---------|---------|---------|---------|
| Bottom-Quantile | 5201    | 5471    | 5521    | 2023    | ..... | 2023    | 3017    | 3033    | 1464    |
| Stock           | 凱衛      | 松翰      | 工信      | 燐輝      | ..... | 燐輝      | 奇鋐      | 威健      | 得力      |
| Top-Quantile    | 2524    | 5432    | 2328    | 5443    | ..... | 4528    | 2524    | 2527    | 2328    |
| Stock           | 京城      | 新門      | 廣宇      | 均豪      | ..... | 江興鍛     | 京城      | 宏璟      | 廣宇      |

Table 20. LightGBM Selection Results

| Date            | 2014/01 | 2014/02 | 2014/03 | 2014/04 | ..... | 2024/11 | 2024/12 | 2025/01 | 2025/02 |
|-----------------|---------|---------|---------|---------|-------|---------|---------|---------|---------|
| Bottom-Quantile | 2451    | 2451    | 4413    | 2451    | ..... | 2451    | 6121    | 2451    | 1805    |
| Stock           | 創見      | 創見      | 飛寶企業    | 創見      | ..... | 創見      | 新普      | 創見      | 寶徠      |
| Top-Quantile    | 2002    | 2538    | 2328    | 2538    | ..... | 2022    | 2023    | 2023    | 2328    |
| Stock           | 中鋼      | 基泰      | 廣宇      | 基泰      | ..... | 聚亨      | 燐輝      | 燐輝      | 廣宇      |

Table 21. XGBoost Selection Results

| Date            | 2014/01 | 2014/02 | 2014/03 | 2014/04 | ..... | 2024/11 | 2024/12 | 2025/01 | 2025/02 |
|-----------------|---------|---------|---------|---------|-------|---------|---------|---------|---------|
| Bottom-Quantile | 4416    | 5523    | 2524    | 1436    | ..... | 2338    | 5212    | 2338    | 6186    |
| Stock           | 三圓      | 豐謙      | 京城      | 華友聯     | ..... | 光罩      | 凌網      | 光罩      | 新潤      |
| Top-Quantile    | 2328    | 2022    | 9904    | 2328    | ..... | 2607    | 1440    | 2539    | 2527    |
| Stock           | 廣宇      | 聚亨      | 寶成      | 廣宇      | ..... | 榮運      | 南紡      | 櫻花建     | 宏璟      |

Table 22. NN1 Selection Results

| Date            | 2014/01 | 2014/02 | 2014/03 | 2014/04 | ..... | 2024/11 | 2024/12 | 2025/01 | 2025/02 |
|-----------------|---------|---------|---------|---------|-------|---------|---------|---------|---------|
| Bottom-Quantile | 5523    | 5523    | 5443    | 3051    | ..... | 5523    | 3051    | 3051    | 3051    |
| Stock           | 豐謙      | 豐謙      | 均豪      | 力特      | ..... | 豐謙      | 力特      | 力特      | 力特      |
| Top-Quantile    | 2328    | 2328    | 5523    | 5523    | ..... | 2328    | 5523    | 5523    | 5523    |
| Stock           | 廣宇      | 廣宇      | 豐謙      | 豐謙      | ..... | 廣宇      | 豐謙      | 豐謙      | 豐謙      |

Table 23. NN2 Selection Results

| Date            | 2014/01 | 2014/02 | 2014/03 | 2014/04 | ..... | 2024/11 | 2024/12 | 2025/01 | 2025/02 |
|-----------------|---------|---------|---------|---------|-------|---------|---------|---------|---------|
| Bottom-Quantile | 6186    | 6186    | 2537    | 2537    | ..... | 2488    | 6192    | 8927    | 2537    |
| Stock           | 新潤      | 新潤      | 聯上發     | 聯上發     | ..... | 漢平      | 巨路      | 北基      | 聯上發     |
| Top-Quantile    | 5523    | 1785    | 2328    | 2328    | ..... | 2537    | 5015    | 2328    | 1785    |
| Stock           | 豐謙      | 光洋科     | 廣宇      | 廣宇      | ..... | 聯上發     | 華祺      | 廣宇      | 光洋科     |

Table 24. NN3 Selection Results

| Date            | 2014/01 | 2014/02 | 2014/03 | 2014/04 | ..... | 2024/11 | 2024/12 | 2025/01 | 2025/02 |
|-----------------|---------|---------|---------|---------|-------|---------|---------|---------|---------|
| Bottom-Quantile | 6128    | 3002    | 5007    | 4109    | ..... | 6128    | 8085    | 8927    | 1737    |
| Stock           | 上福      | 歐格      | 三星      | 加捷生醫    | ..... | 上福      | 福華      | 北基      | 臺鹽      |
| Top-Quantile    | 6242    | 2538    | 2328    | 2328    | ..... | 3060    | 5355    | 2328    | 6285    |
| Stock           | 立康      | 基泰      | 廣宇      | 廣宇      | ..... | 銘異      | 佳總      | 廣宇      | 啟基      |

Table 25. NN4 Selection Results

| Date            | 2014/01 | 2014/02 | 2014/03 | 2014/04 | ..... | 2024/11 | 2024/12 | 2025/01 | 2025/02 |
|-----------------|---------|---------|---------|---------|-------|---------|---------|---------|---------|
| Bottom-Quantile | 6242    | 6242    | 8927    | 8927    | ..... | 6242    | 8927    | 6242    | 8050    |
| Stock           | 立康      | 立康      | 北基      | 北基      | ..... | 立康      | 北基      | 立康      | 廣積      |
| Top-Quantile    | 2504    | 4205    | 2328    | 2328    | ..... | 4205    | 2328    | 8927    | 4906    |
| Stock           | 國產      | 中華食     | 廣宇      | 廣宇      | ..... | 中華食     | 廣宇      | 北基      | 正文      |

Table 26. NN5 Selection Results

| Date            | 2014/01 | 2014/02 | 2014/03 | 2014/04 | ..... | 2024/11 | 2024/12 | 2025/01 | 2025/02 |
|-----------------|---------|---------|---------|---------|-------|---------|---------|---------|---------|
| Bottom-Quantile | 2031    | 6291    | 3232    | 2538    | ..... | 6291    | 2538    | 2538    | 8074    |
| Stock           | 新光鋼     | 沛亨      | 昱捷      | 基泰      | ..... | 沛亨      | 基泰      | 基泰      | 鉅橡      |
| Top-Quantile    | 8112    | 2328    | 3049    | 2328    | ..... | 2434    | 2328    | 2328    | 8112    |
| Stock           | 至上      | 廣宇      | 精金      | 廣宇      | ..... | 統懋      | 廣宇      | 廣宇      | 至上      |

*Note:* All figures and tables presented in this study are produced by the authors based on the Taiwan Economic Journal (TEJ) dataset described above, unless otherwise indicated in the caption.

## **7 Conclusion**

This study uses monthly data of listed companies in Taiwan from January 1999 to February 2025, focusing on five factors: short-term momentum, long-term momentum, momentum change, maximum single-period return, and return volatility to construct a multi-level factor dataset. Five machine learning models (Random Forest, Neural Network, Gradient Boosting Regression Tree, LightGBM, XGBoost) and linear regression are employed to model and predict the relationship between these factors and future returns. Based on the predicted returns, stocks are ranked in equal partitions to construct investment portfolios, which are then compared with the Taiwan Weighted Stock Index (TWII) for performance evaluation.

Empirical results indicate that the five nonlinear machine learning models consistently outperform linear regression, particularly in capturing the nonlinear relationships between momentum factors and extreme returns, effectively improving portfolio winning rates and cumulative returns. Equal-partition ranking also allows flexible adaptation to different investment strategies, enabling diversified asset allocation and risk management. Overall, machine learning methods not only enhance the accuracy of factor prediction and stock selection but also provide a practical empirical framework for intelligent investing.

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